OPERATING MODES, SAMPLING ARCHITECTURE, AND POWER MANAGEMENT

Adaptive Operational Concept

The longitudinal penile physiology monitoring device may operate under multiple configurable operating modes designed to balance physiologic measurement fidelity, computational workload, and energy consumption. Operational modes may be manually selected by a user, automatically determined by device algorithms, scheduled according to time-of-day or behavioral patterns, or dynamically adjusted based on detected physiologic or motion conditions.

Sampling architecture may be configured to support continuous acquisition, periodic sampling, burst acquisition, event-triggered sampling, or hybrid acquisition strategies. Sampling control may be applied independently across sensing modalities such that rigidity sensing, temperature monitoring, inertial measurement, and optical perfusion sensing may operate at different temporal resolutions simultaneously.

Performance-Oriented Operating Mode

A performance or high-fidelity operating mode may enable elevated sampling rates, increased sensor activation frequency, expanded signal processing operations, and higher-resolution physiologic characterization. This mode may prioritize detailed analysis of erection onset dynamics, rigidity stability, vascular response behavior, or short-duration physiologic events at the expense of reduced battery longevity.

Balanced Monitoring Mode

A balanced operating mode may provide intermediate sampling rates and processing intensity to maintain meaningful longitudinal data collection while preserving practical battery life. Balanced operation may combine continuous low-rate monitoring with periodic higher-rate sampling intervals triggered by physiologic or motion indicators.

Longevity-Focused Operating Mode

A longevity mode may minimize energy consumption through aggressive duty cycling, reduced sensor activation, and event-driven data acquisition. Under longevity operation, baseline monitoring may occur at low sampling rates while detected physiologic deviations automatically trigger temporary high-fidelity acquisition windows to preserve critical information without continuous energy demand.

Adaptive Sampling and Event Detection

Adaptive algorithms may evaluate incoming sensor data to identify physiologic transitions such as tumescence onset, rigidity stabilization, detumescence, or motion state changes. Detection of such events may dynamically modify sampling parameters, activate additional sensing channels, or initiate temporary computational processing modes. Event detection may improve longitudinal characterization while maintaining efficient power utilization.

Power System and Energy Management

The device may include rechargeable or replaceable energy storage components optimized for minimal device profile and extended wear. Power management strategies may include low-power microcontroller operation, sensor duty cycling, dynamic voltage scaling, sleep states, wake-on-event circuitry, wireless communication scheduling, and predictive energy budgeting. Charging may be accomplished through electrical contacts, magnetic coupling, inductive charging, or alternative energy transfer methods.

Operational architectures described herein are illustrative and non-limiting and encompass future power management techniques and adaptive sampling technologies capable of enabling efficient longitudinal physiologic monitoring.